Item 12: a surrogate for the solving-degree surface

Verdict: MATCHES (primitive), no speedup, no gap to rho

Under the frozen rule in `PROTOCOL.md`, the gradient-boosted surrogate (arm c1) is **not separated from the "levelled" engine-median reading (arm b)**. It does beat the semi-regular (Froberg) reading (arm a) by a wide margin. Its 90 % conformal intervals cover 73.5 % of held-out draws, outside the [0.85, 0.95] band, and its error is inside the label-permutation null. Level: **primitive** (a planning instrument). Nothing here is an exponent, a constant or a speedup; no `S`, rho ratio or GAE is computed, so the section 5a 0 / 0.1 / 1 GAE pricing band does not apply and the **gap to matched rho is not applicable** (no cross-method row exists). Result is a stage diagnostic and cannot promote a complete-DLP claim. Not a performance result, so nothing is copied to `benchmarks/`.

Question and method

Can a model fitted on committed toy-cell degree records predict the solving degree `D` of cells it has not seen better than (a) the semi-regular `D_reg` and (b) the engine's median `D` ("levelled")? No new algebraic measurement: 404 records (345 exact, 59 right-censored), 8 families, harvested from 194 committed files each pinned by sha256 in `sources.tsv` (the tool refuses a mismatch; `SOURCES_VERIFIED 194`). `research/ffd_*` does not exist and `research/f6_*` holds no solving degree; items 04, 05, 07 were unfinished and are not used. Arms: (a) `D_reg`, (b) engine median, (c1) boosted trees on shape covariates (depth 3, 200 rounds, rate 0.1), (c2) c1 plus `D_reg` (ablation), (e) ridge. Evaluation: leave-one-family-out over 7 evaluable families, training rows whose `(m, n, l)` shape occurs in the held-out family removed (shape-disjoint), 20 split-conformal repeats, macro mean over folds, 2000-resample cluster bootstrap (paired), 100 within-engine label permutations.

Numbers (measured; `raw/eval.kv`, `raw/null.kv`, `raw/verdict.txt`)

arm	macro MAE of D	macro MAE of log2 size	90 % cover	Spearman rho
a Froberg `D_reg`	2.382	4.039	0.914	0.661
b engine median	**0.391**	**0.355**	0.954	0.023
c1 boosted trees (primary)	0.482	0.538	**0.735**	0.519
c2 c1 + `D_reg`	0.494	0.539	0.740	0.542
e ridge	1.032	1.406	0.389	-0.044

Paired bootstrap 95 % intervals (positive = c1 better):

• MAE(a) - MAE(c1) = [1.647, 2.137]: c1 beats a.

• MAE(b) - MAE(c1) = [-0.195, 0.006]: contains 0; c1 does not beat b (point estimate favours b by 0.09). On log2 size, b - c1 = [-0.340, -0.039], b is better (outside the rule).

• MAE(c2) - MAE(c1) = [-0.011, 0.031]: `D_reg` as a feature adds nothing.

• MAE(e) - MAE(c1) = [0.399, 0.695]: trees beat the linear baseline.

Per fold (`raw/evaluate.txt`, MAE of D, a / b / c1): cl-m3 2.60 / 0.07 / 0.54; cl-m4 4.04 / 0.18 / 0.27; f4-m3 3.00 / 0.00 / 0.05; f4-m4 3.54 / 0.23 / 0.28; xl-m2 2.75 / 1.10 / 1.09; xl-m3 0.33 / 0.72 / 0.62; xl-m4 0.43 / 0.44 / 0.53. Froberg is best on xl-m3 and xl-m4 (where `D` grows with the shape) and wrong by 2.6 to 4.0 degrees on the closure and F4 families (where `D` sits at 3 to 4); the engine median is best where `D` is constant; c1 sits between and wins outright only on xl-m2 and xl-m3 by margins under its spread.

Null: the real c1 macro MAE over the same 3 splits is 0.569 against a null band of 0.483 to 0.575 (median 0.523; 93 of 100 permutations at or below the real value). The real error is inside the band, so the rule does not count c1 as a signal beyond the engine marginal.

Censored consistency (descriptive, 59 records): share with `y_hat + q >= bound`: a 1.00, b 0.96, c1 0.66, c2 0.66, e 0.64. Intervals of c1 often sit below degrees the censored runs already exceeded.

Sensitivity without the shape-disjoint rule (outside the verdict, `leaky_*`): a 2.382, b 0.379, c1 0.430, c2 0.439, e 0.490. Removing the rule lets shapes leak across engines and the surrogate gets closer to b but still does not beat it.

T3 (descriptive, outside the rule; 61 finished F4 runs, folds f4-m3 and f4-m4): MAE of log2(columns) a 7.10, b 2.87, c1 2.25, e 2.39; rho c1 0.20. Two folds only; no conclusion.

Controls (all passed)

• Known answers: 13 `D_reg` values and the 124,314-column count at `N = 42, d = 4`; 14 selftest checks (`raw/selftest.kv`, `selftest_ok=1`).

• Conformal on i.i.d. synthetic data: coverage 0.8997 over 200 repeats.

• Positive control (planted label `3 + l + [S < 0]` on the real covariates): c1 MAE 0.629 against b 1.938, passes `c1 <= 0.5 * b`. The pipeline can see an effect that exists.

• Negative control: the permutation null above. It is the reason the verdict is MATCHES and not ADVANCES: c1's error is what a label-shuffled model attains.

The five cells to buy (model output, not a measurement; `raw/rank.txt`)

rank	engine	m	n	l	N	c1 prediction	`D_reg`	predicted log2 columns
1	cl	4	14	7	56	4.13	14	18.6
2	cl	4	14	6	52	4.05	13	18.2
3	cl	4	19	5	58	4.07	13	18.8
4	f4	4	22	3	56	3.85	12	18.6
5	cl	4	22	4	60	3.88	12	19.0

Because c1 is not separated from the engine median, its prediction is about 4 everywhere, so the ranking reduces to "largest `D_reg`". Read it as the cells where the "levelled" and "growing" readings disagree most, which one measurement would settle, not as a model-derived expectation of `D`.

Deviations (all tightenings or clarifications; `PROTOCOL.md`)

Arm (c) pinned to c1; bootstrap, permutation null and controls added to the design's rule; macro averaging; censored labels excluded from training and error; shape-disjoint folds; `rustc` build per card instead of `cargo build --example`. Protocol commit e923d2792, amended in 3244112ed and b88ad9129 before any measured run. The card relays were saved as `raw/relay-1.txt` to `relay-3.txt`; `tools/task_card.py check-relay` reports `relay_verbatim: true` for all three. The commit hashes in receipts 02 and 03 (26916471b, 77395371a) are later HEADs from other lanes' commits, not changes to this item's inputs (card sha256 values match).

Caveats

• Host is L0 (shared 4-vCPU container, no isolation). Nothing here is a timing.

• Toy scale: n up to 24, `l` up to 7, `D` between 3 and 8. No `n = 131` claim.

• Labels are mostly constant within an engine (cl and f4 at 3 to 4), so b is a strong baseline and the low-variance folds reward it. f4-m3 has only 4 cells.

• The evaluable folds are different engines and `m` values, so every fold is a covariate shift; the nominal 90 % was a hypothesis and it failed (0.735).

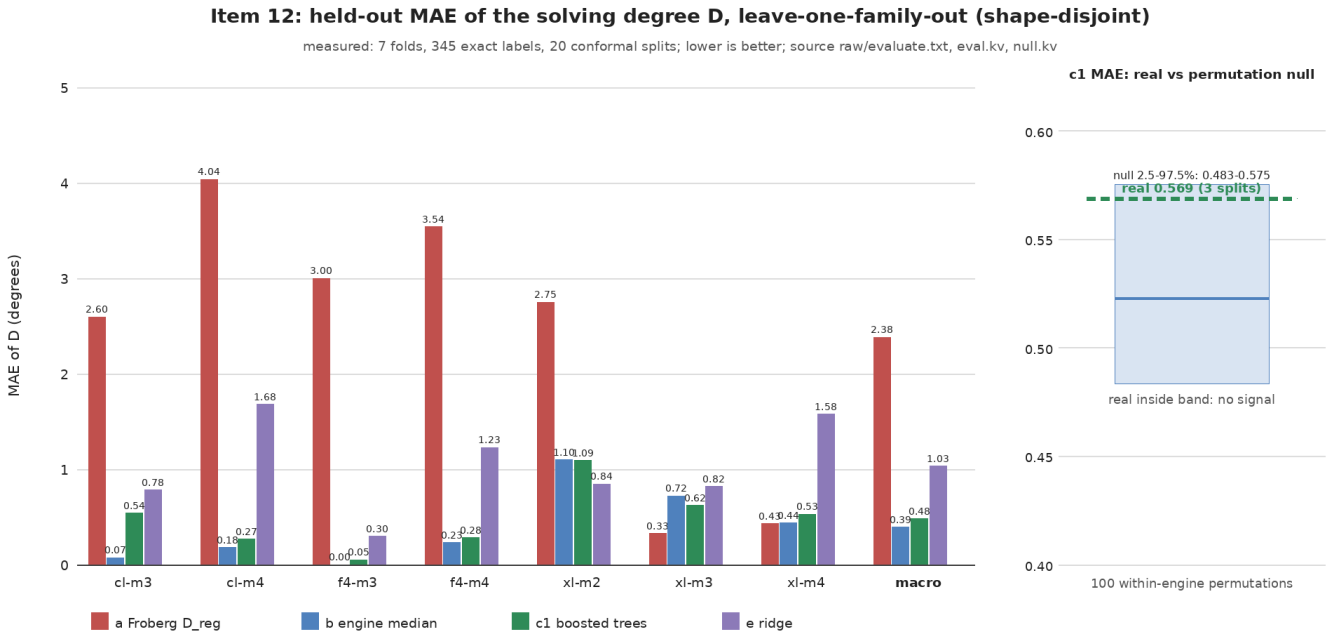
• Items 04, 05, 07 cells are a prospective test set for a later addendum that cannot change this verdict.

• The Conductor CLI is not installed on this host; no `conductor check` was possible. Edits were confined to this item's files.

Files

`PROTOCOL.md`, `card-01..03.yaml`, `sources.tsv`, `raw/` (data, kv, receipts, relays), `report/graph.svg`, `report/graph.png`, `report/graph-data.json`, `report/plot.py` (layout only), `REPORT.pdf`.

Graph: held-out MAE of D per fold, and the permutation null



Caption: CARD-BLINDSPOT-12-02/03, commit b88ad9129 protocol; verdict MATCHES (primitive). Toy-scale labels, host L0 (shared 4-vCPU). Bars clipped at 5.
Arm a is the semi-regular reading; arm b the engine-median "levelled" reading; c1 is not separated from b (diff b-c1 95% CI [-0.195, 0.006]).

Source: raw/evaluate.txt, raw/eval.kv, raw/null.kv (CARD-BLINDSPOT-12-02). Plotted values only; see report/graph-data.json.